Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/21**

Paper 2 Theory **May/June 2023**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)	CaO (1)	**1**
1(b)	Al_2O_3 (1)	**1**
1(c)	SiO_2 (1)	**1**
1(d)	CaO (1)	**1**
1(e)	H_2O (1)	**1**
1(f)	CO_2 (1)	**1**

Question	Answer	Marks
2(a)	**Any two from:** soft (1) low density / floats on water (1) low melting point (1) good thermal conductor / conducts heat (1) good electrical conductor / conducts electricity (1) lustrous / shiny (when freshly cut) (1) ductile (1) malleable (1)	**2**
2(b)(i)	potassium loses electrons	**1**
2(b)(ii)	0 (1)	**1**

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Question	Answer	Marks
2(b)(iii)	**Any three from:** (potassium) floats on water / moves across the surface of water (1) (potassium) melts (1) bubbles / fizzes / effervesces (1) forms a colourless solution / piece gets smaller and smaller (1) lilac flame (1)	**3**
2(c)	yellow / orange (1)	**1**

Question	Answer	Marks
3(a)(i)	(moles of zinc oxide =) 3.50 / 81 **OR** 0.0432 (1) (moles of HCl =) 0.0600 (1) idea that need 0.0864 moles of HCl / only 0.0300 moles of ZnO react (1)	**3**
3(a)(ii)	so **all** the **acid** is used up (1)	**1**
3(a)(iii)	filtration (1)	**1**
3(b)	barium nitrate or barium chloride **AND** any soluble sulfate (1)	**1**
3(c)(i)	nitric acid **and** sodium hydroxide (1)	**1**
3(c)(ii)	titration (1)	**1**

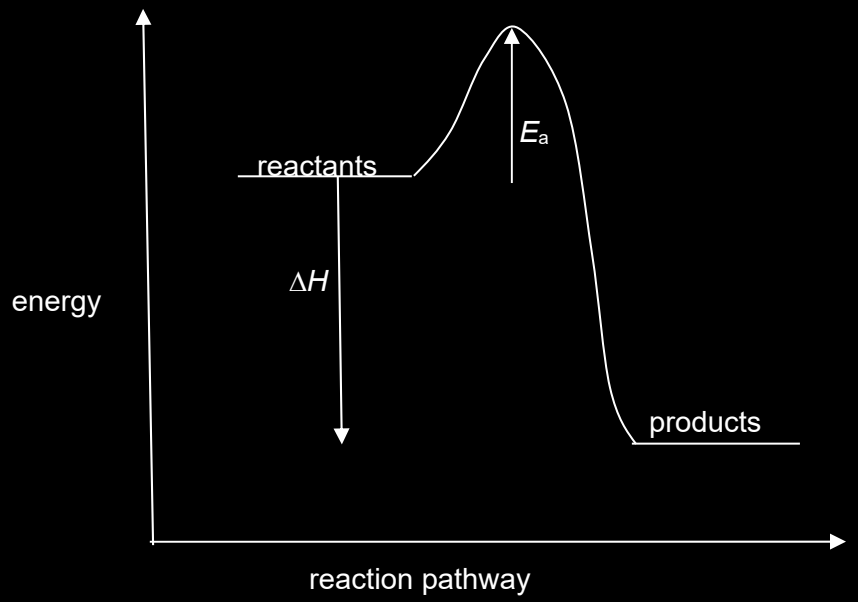
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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>4(a)</td><td>
<table>
<tr><td>particle</td><td>number of particles</td></tr>
<tr><td>electron</td><td>18 (1)</td></tr>
<tr><td>neutron</td><td>16 (1)</td></tr>
<tr><td>proton</td><td>15 (1)</td></tr>
</table>
</td><td>3</td></tr>
<tr><td>4(b)</td><td>P^{3-} has full outer shell of electrons (1)</td><td>1</td></tr>
<tr><td>4(c)</td><td>Ca_3P_2 (1)</td><td>1</td></tr>
<tr><td>4(d)</td><td>giant structure (1)
strong attraction between positive and negative ions / strong electrostatic attraction between ions (1)</td><td>2</td></tr>
<tr><td>4(e)</td><td>M_r = 310 (1)
% = 20% (1)</td><td>2</td></tr>
</table>

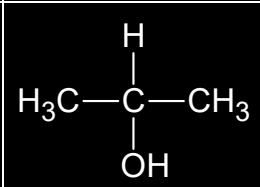
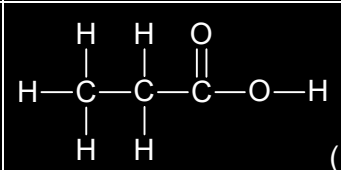
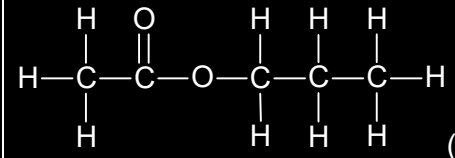
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Question	Answer	Marks
5(a)(i)	(a reversible reaction) in which the **rate** of the forward reaction is the same as the **rate** of the backward reaction (1) and the concentrations of the reactants and the products do not change / are constant (1)	**2**
5(a)(ii)	(position of equilibrium) moves to the right (1) because the reaction is endothermic / to minimise temperature increase by absorbing energy (1)	**2**
5(a)(iii)	position of equilibrium moves to the left (1) because more moles of **gas** on product side / fewer moles of **gas** on reactant side (1)	**2**
5(b)	rate increases because particles move faster / particles have more kinetic energy (1) more successful collisions / more collisions involve particles with equal or more than activation energy (1)	**2**
5(c)	rate increases because more particles per unit volume (1) greater collision frequency (1)	

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Question	Answer	Marks
6(a)	 correct labels for axes (1) reactants on a line (1) product on a line to the right **and** below reactant line (1) vertical line with arrow and correct label for enthalpy change (1) vertical line with arrow and correct label for activation energy (1)	**5**
6(b)	energy associated with bond breaking = 2 × 298 **OR** 596 (1) energy associated with bond forming = 436 + 151 **OR** 587 (1) (enthalpy change = 596–587) = +9 (kJ/mol) (1)	**3**

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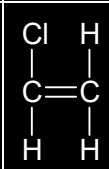
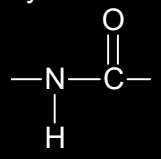
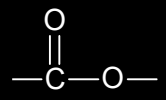
Question	Answer	Marks
7(a)	$C_nH_{2n+1}OH$ (1)	**1**
7(b)(i)	(structural) isomers (1)	**1**
7(b)(ii)		**1**
7(c)	contains **only** single bonds between carbon atoms (1)	**1**
7(d)	contains oxygen (as well as carbon and hydrogen) (1)	**1**
7(e)(i)	 (1)	**1**
7(e)(ii)	 (1)	**1**
7(f)(i)	3 C–H bonds and one C–O bond around carbon (1) rest of structure correct (1)	**2**
7(f)(ii)	electrons cannot move / all electrons are in bonds	**1**
7(g)	(liquid that) **dissolves** a solute (1)	**1**

Question	Answer	Marks
8(a)	KC*l* (aq) oxygen at anode and hydrogen at cathode $CuSO_4$(aq) oxygen at anode and copper at cathode PbI_2(l) iodine at anode and lead at cathode **two** or **three** correct products = 1 mark **four** or **five** correct products = 2 marks **all six** products = 3 marks	**3**
8(b)	**Any two from:** electrolyte stays blue (1) cathode gets coated with a pink solid (1) anode gets smaller (1)	**2**
8(c)	$2Cl^- \rightarrow Cl_2 + 2e^-$ / $2Cl^- - 2e^- \rightarrow Cl_2$	**1**

Question	Answer	Marks
9(a)	volume of NO = (0.002 × 960) ÷ 100 **OR** 0.0192 (1) moles of NO = (0.0192 ÷ 24) **OR** 0.0008 (1) number of molecules = $0.0008 \times 6.02 \times 10^{23} = 4.816 \times 10^{20}$ (1)	**3**
9(b)	$N_2 + O_2 \rightarrow 2NO$ (1)	**1**
9(c)	acid rain / photochemical smog / respiratory problems (1)	**1**
9(d)	reacted with CO to make nitrogen (1)	**1**
9(e)(i)	NO_2 has a higher (relative) molecular mass / molecule has more mass (1)	**1**

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Question	Answer	Marks
9(e)(ii)	(NO) molecules move slower / (NO) molecules have less **kinetic** energy (1)	**1**

Question	Answer	Marks
10(a)	 (1)	**1**
10(b)(i)	carbon monoxide (1)	**1**
10(b)(ii)	land-fills (are filled) / accumulation of plastics in oceans (1) chemically unreactive / do not dissolve in water (1)	**2**
10(c)	nylon / protein / polyamide / polyester / PET corresponding linkage e.g.  or 	**2**

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